

Transformer Inference Cost Analysis, Sequential Data

Full-Sequence vs. RMT-Style Blocking strategy Inference for Protein Fitness Prediction

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APAI 3 CFU project work

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Preliminary

Protein Sequences

- Protein in their primary structure (sequence), can actually be seen as a sequential data, similar to text, but with only 20 alphabets (amino acids)
- Much shorter than typical NLP inputs, longest sequences in famous databases are around 3000-4000 residues, so long-range dependency here isn't as critical as it is in NLP
- A good fit for this project: modest compute needs, plenty of public benchmark data, and block-splitting strategies for proteins are still an open research question, motivating a cost vs accuracy-drop analysis

ESM model

- This is a pretrained protein language model, trained on a massive amount of data from a famous biology database called UniProt
- Introduced by Facebook research (currently Meta AI), and it has several versions and variants
- I have selected 2 models from this family (1 medium sized: ESM-2 35M, and 1 large size: ESM-2 150M), to compare them within my explorations

ProteinGym database

- This is a collection of benchmarks aiming at comparing the ability of models to predict the effects of protein mutations
- It's built around Deep Mutational Scanning (DMS) experiments (Something conceptually similar to RLHF in the NLP world!)

Preliminary

The selected data from ProteinGym

Assay	Bucket	Length
SQSTM_MOUSE_Tsuboyama_2023_2RRU	short	40
SUMO1_HUMAN_Weile_2017	short	101
TRPC_SACS2_Chan_2017	medium	248
SPG1_STRSG_Wu_2016	medium	448
A4_HUMAN_Seuma_2022	long	770
A0A140D2T1_ZIKV_Sourisseau_2019	long	3423

Number of sequences in each category is also the same. Which I Think is a fair point for the project to do the cost analysis based on different important parameters including the input size.

Motivation and Goal

The problem

- ESM-2 uses full self-attention, cost grows quadratically with sequence length: $O(n^2)$
- Long proteins become disproportionately expensive. Beyond some length, the model does not fit in GPU memory. We will see in the next slides! :))

The idea

- Retrofit a pretrained ESM-2 checkpoint with an RMT-style inference procedure, no retraining, only do changes in inference time
- Split the sequence into **fixed-size** blocks. **Carry learned memory tokens across blocks**
- Goal: convert quadratic cost into linear cost

Two-part investigation

- Part 1: theoretical, instruction-by-instruction cost analysis
- Part 2: empirical cost and accuracy analysis on real ProteinGym sequences

RMT vs. Alternatives

Selected: RMT-style segment-recurrent inference

- Backbone (ESM-2) untouched, a bolt-on wrapper is added around it
 - vs. full-sequence attention: exact, accurate, but $O(n^2)$
 - vs. Block-Recurrent Transformer: more expressive, but needs a custom cell trained from scratch, out of scope for 3 CFU project work
 - vs. Universal Transformer: recurrence over depth, not sequence length, different point of view. Based on what motivates me to do this project, I'm more interested in the input size analysis rather than the network depth, which is also nice vision, but different from the project aspects
 - vs. sparse attention: needs a custom kernel or fine-tuning, RMT is a cleaner less heavy approach worths giving it a shot!

Pros

- Retrofittable
 - the recurrent memory mechanism can be added to an existing, pre-trained Transformer model without altering its core internal architecture
- memory independent of sequence length
- simple to implement

Cons

- Accuracy drop (losing information when we try to compact all previous block infos in m memory tokens)
- sequential across blocks, not embarrassingly parallel.
- No Biological inspiration
 - the splitting method could be inspired by biological aspects to make more sense, and potentially, reach better accuracy. Even if I test all the combinations, I will lose the interpretability of why some sorts of splittings work better than others

Theoretical Cost Comparison

	Full-sequence attention	RMT block-recurrent
Time (FLOPs)	$O(L(n^2d + nd^2))$	$O(L(nbd + nd^2)) \rightarrow O(n)$
Peak memory	$O(n^2)$	$O(b^2)$ — independent of n
Parallelism	one pass, fully parallel	s sequential passes
Accuracy	exact global context	lossy, m -token bottleneck

- n = sequence length, b = block size, m = memory tokens, $s = \lceil n/b \rceil$ segments — d , b , L treated as fixed constants
- d : the hidden dimension of the transformer, the size of each token's embedding vector inside ESM-2 (35M): 480. ESM-2 (150M): 640
- Ideal vs. effective: RMT carries a fixed $\approx m/b$ tax; the quadratic term only dominates once n is large relative to d

Methodology

Models

- esm2_t12_35M_UR50D (35M) and esm2_t30_150M_UR50D (150M)

Dataset

- ProteinGym DMS substitution benchmark: 6 assays, stratified short / medium / long (40–3423 residues)
- Same real sequences used for both cost and accuracy measurement

Cost measurement

- MACs via fvcare
- peak GPU memory via torch.cuda: real sequences, $b \in \{128, 64, 32, 16\}$
 - `torch.cuda.max_memory_allocated() / 1024**2` -> to compute the memory in MB instead of bytes

Accuracy measurement

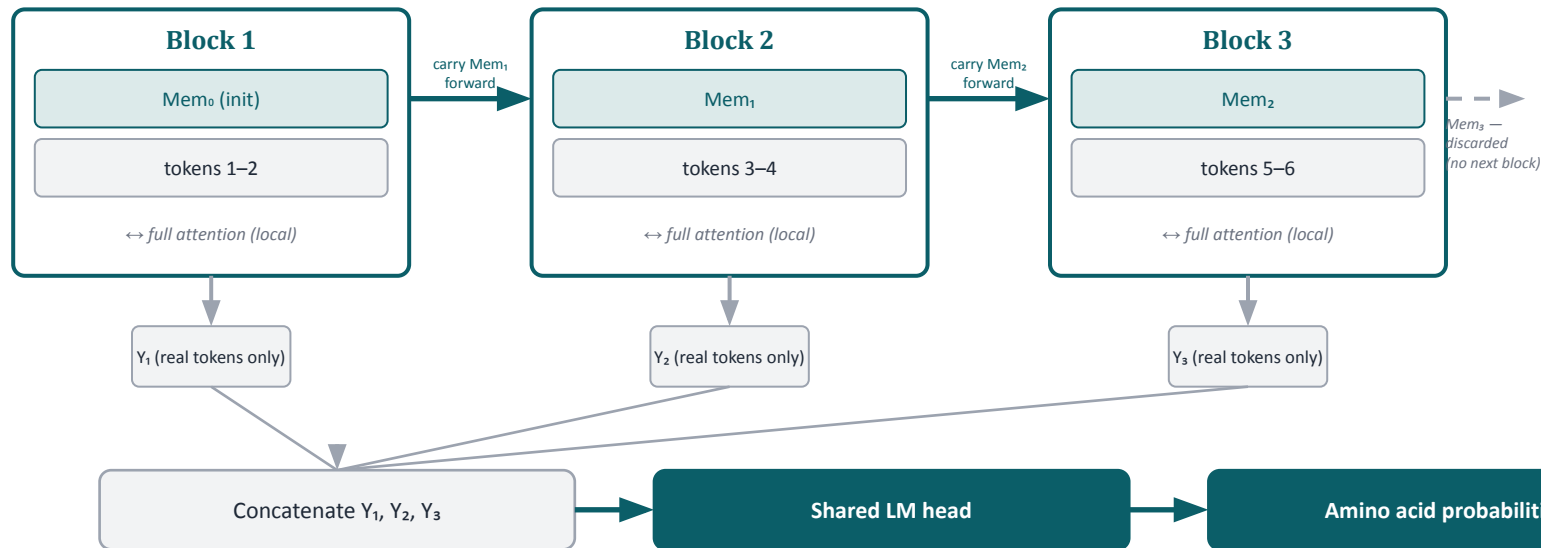
- Masked-marginal zero-shot scoring, correlated (Spearman ρ) against experimental DMS scores
 - `score = log P(mutant amino acid) - log P(wildtype amino acid)`
- Do the model's predicted scores **rank** mutations in the same order as the real experimental measurements did?

Ablation

- Isolates memory-token count (m) and recurrence (carry_memory: True vs. False)

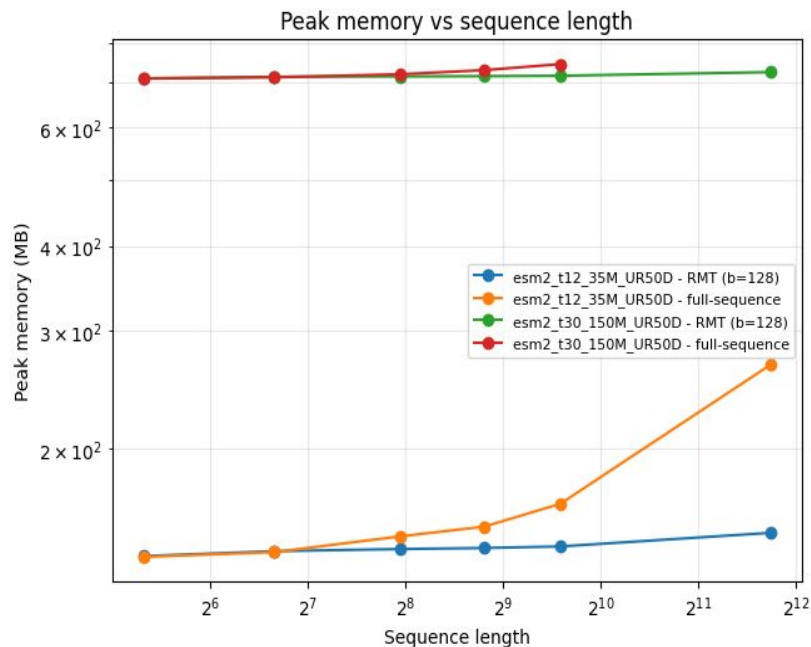
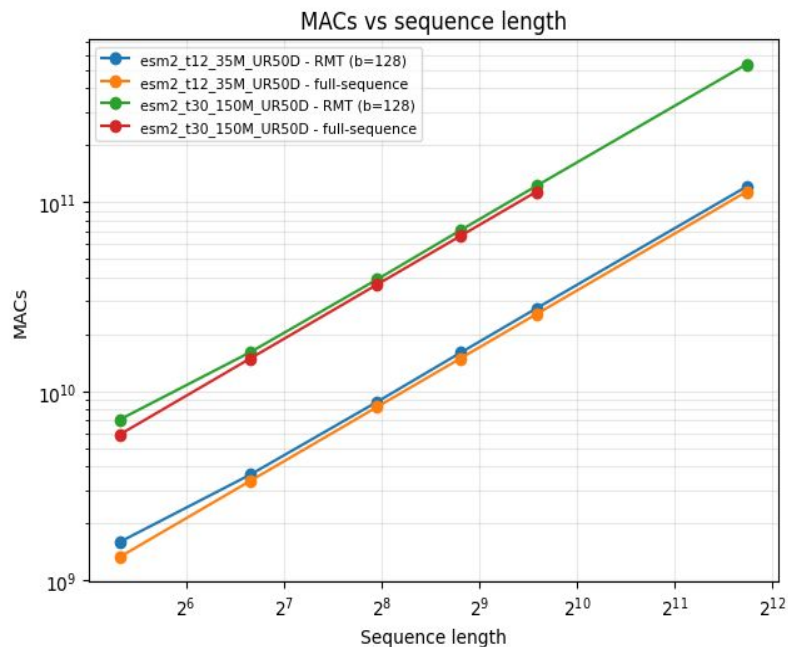
Recurrent Memory Transformer (RMT) Algorithm

How Blocks Combine via Memory Tokens



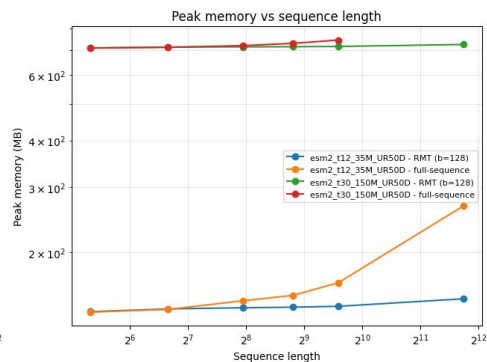
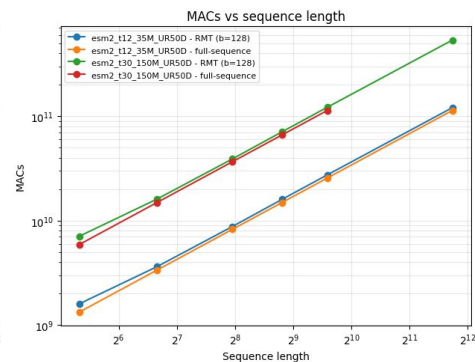
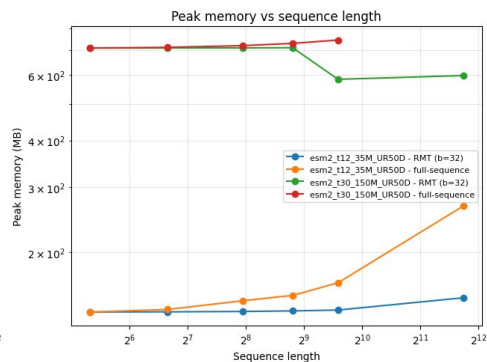
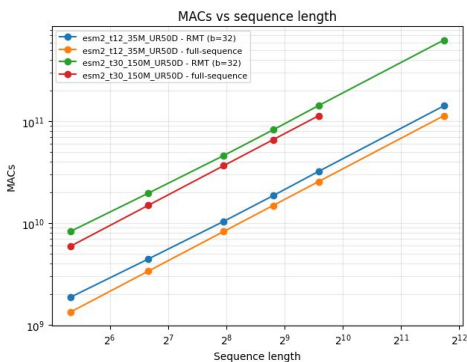
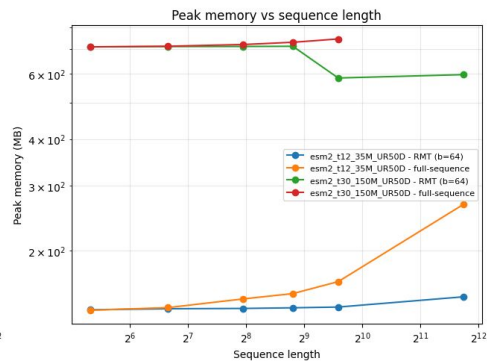
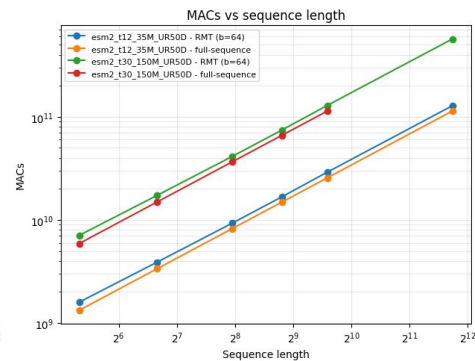
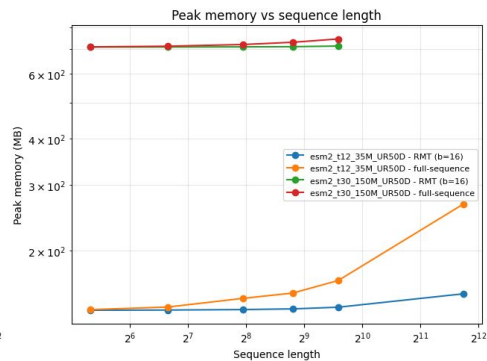
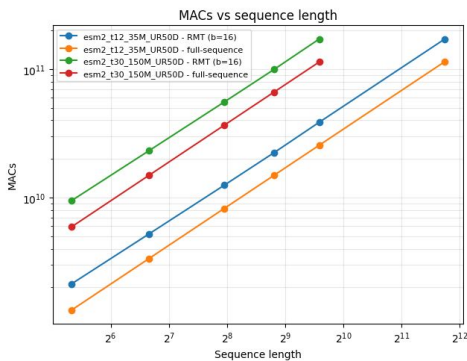
- Blocks combine through re-injection of memory vectors, never by merging attention scores across blocks
- This is the mechanical source of the accuracy bottleneck: real tokens carry their full d-dim representation forward; everything crossing a block boundary must fit through just m memory slots
- $m \ll b$

Results: Cost analysis based on Sequence length

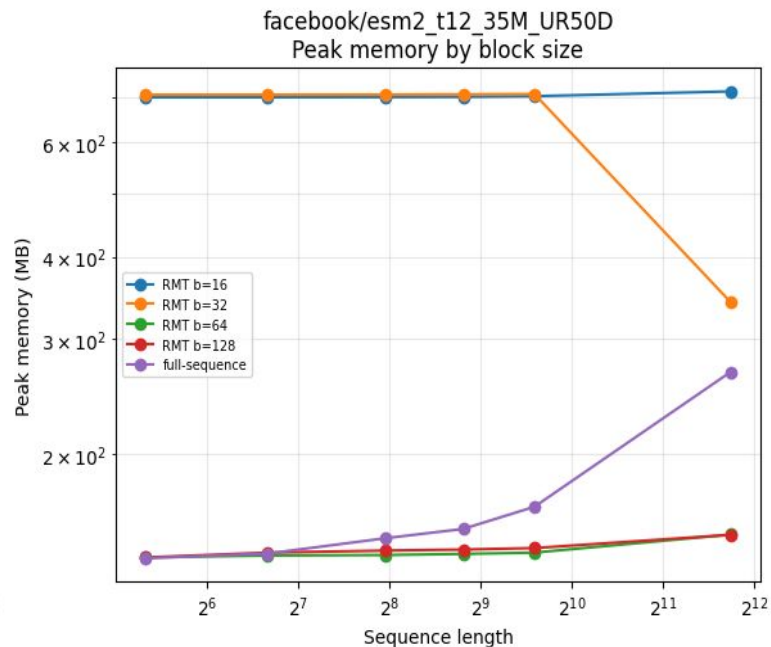
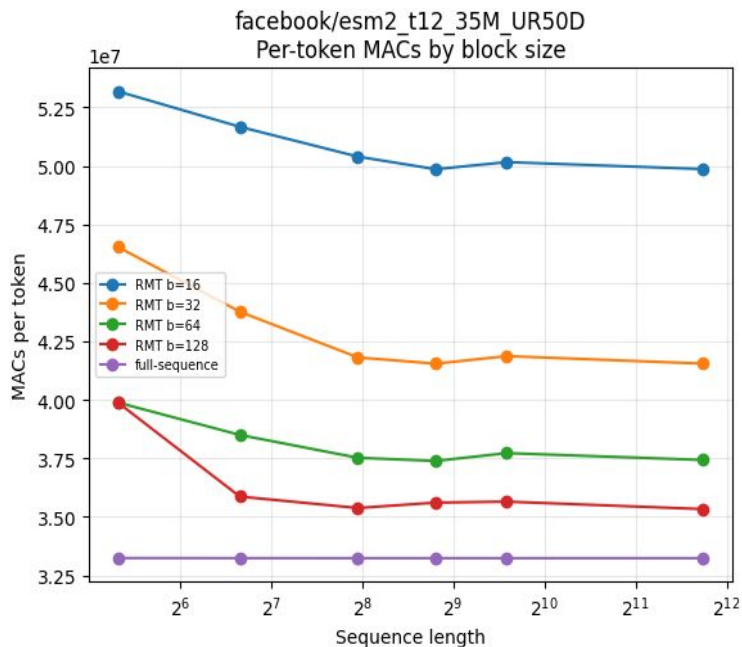


Both models, block size $b = 128$. Note the missing 150M full-sequence point at $n = 3423$ (OOM).

Results: Cost analysis based on Sequence length

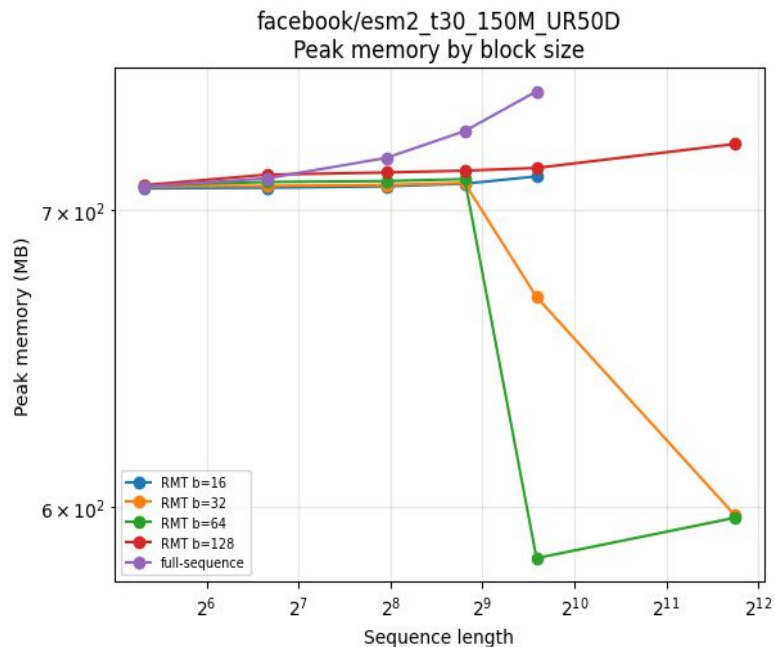
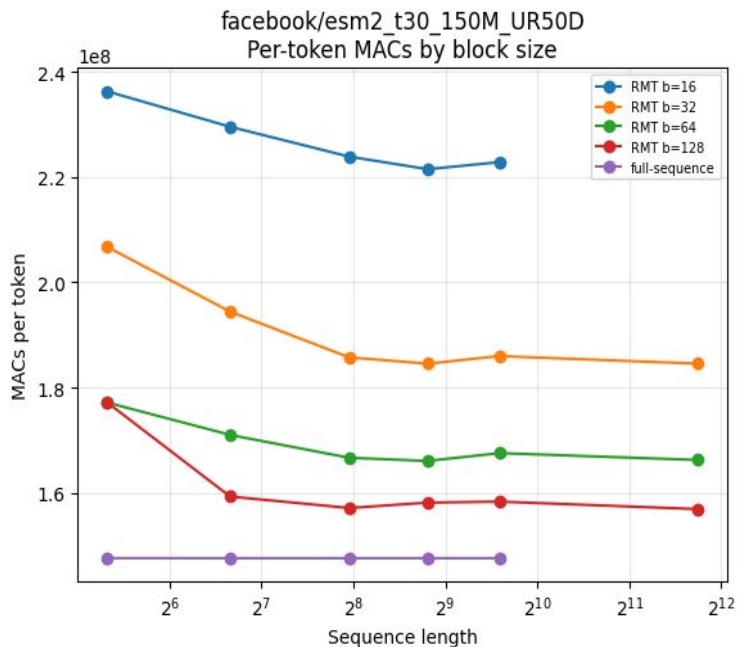


Effect of Block Size on Cost, 35M Model



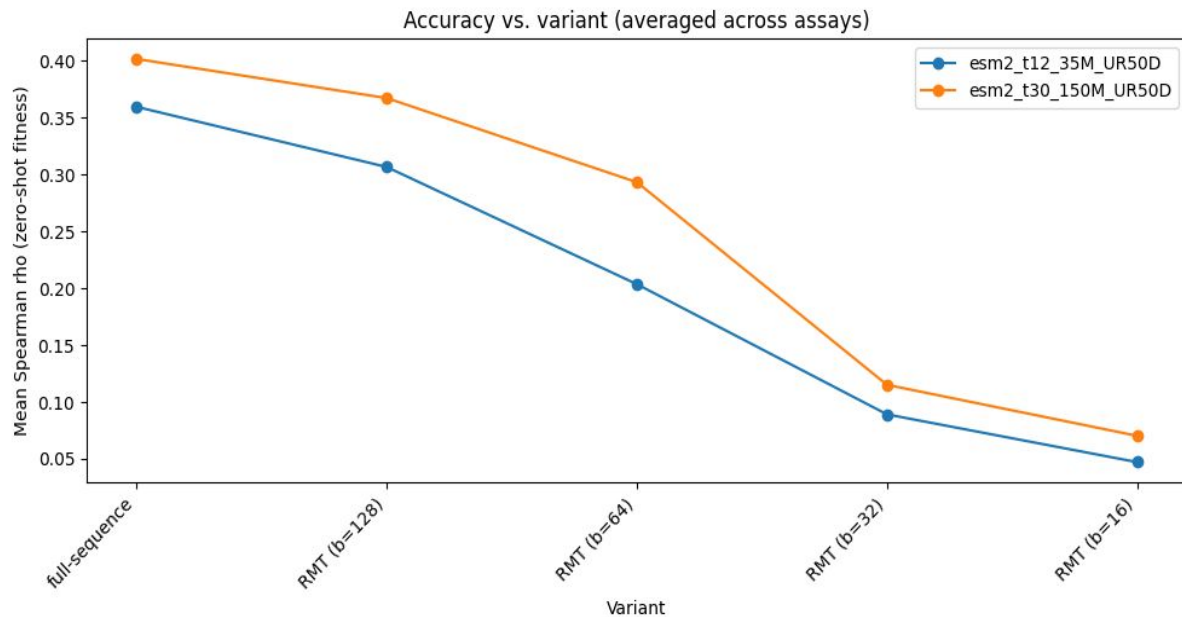
Smaller blocks: higher per-token MACs (fixed m/b tax), flatter memory growth.

Effect of Block Size on Cost, 150M Model



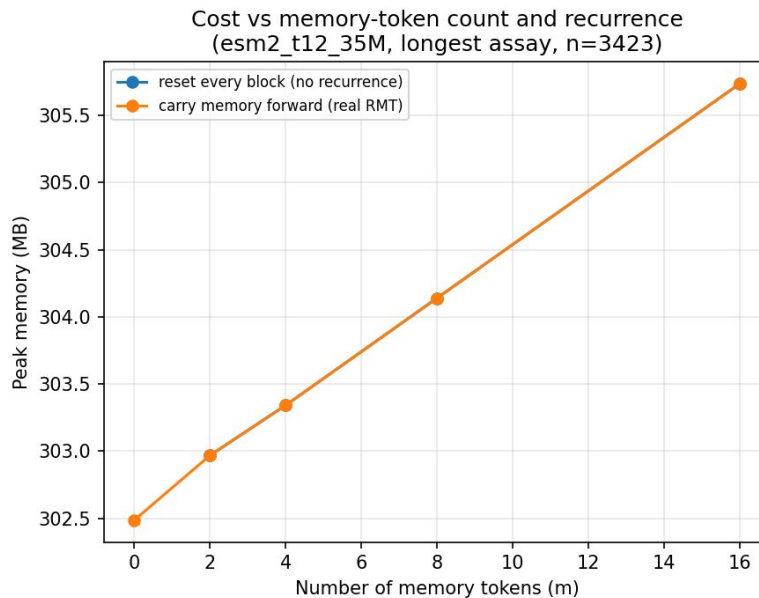
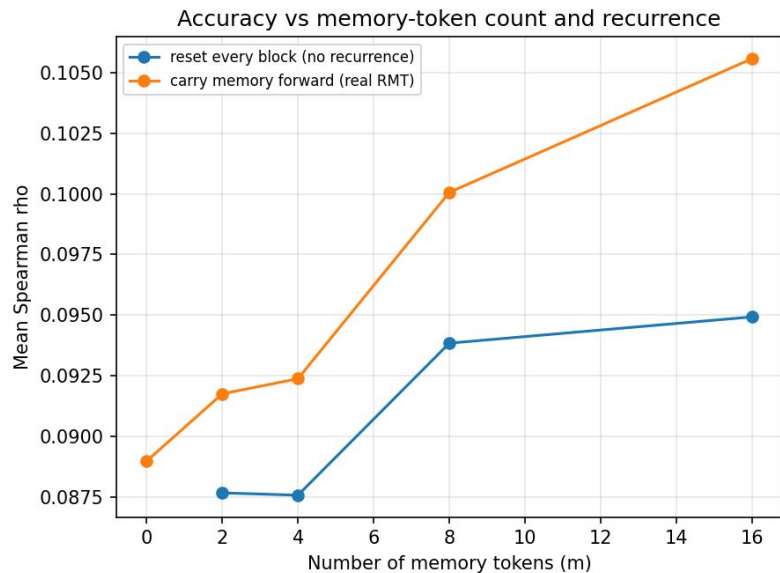
$n = 3423$ is OOM for RMT $b=16$, and for the full-sequence

Results: Accuracy Across Variants



Mean Spearman ρ across all 6 assays — monotonic degradation as block size shrinks. At $b=128$, RMT on short proteins nearly matches full-sequence.

Ablation Study: Memory Tokens and Recurrence



*carry_memory=True beats False at every $m > 0$, peak memory is identical either way (which was expected!)
Mean Spearman rho: averaging rhos for each essay together*

Conclusion

- Theory predicts $O(n^2)$ time & $O(n^2)$ memory (full) vs. $O(n)$ time & $O(b^2)$ memory (RMT)
- Memory side confirmed empirically, a real, decisive OOM boundary crossed only by full-sequence
- Compute side not yet very visible at these sequence lengths
- Accuracy degrades smoothly and predictably as block size shrinks
- Ablation: recurrence does real work, and costs nothing extra in memory
- RMT is not simply better or worse, it is a specific, measured point on a memory–accuracy trade-off curve

Thank you for your attention